MIL-R-61001
30 March 1989
SUPERSEDING
AA-R-0045 (GSA-FSS)
27 January 1969
(See 6.4)

MILITARY SPECIFICATION

RACKS, STORAGE, LOADED PALLET

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers boltless, single and multiple opening, steel and galvanized steel pallet storage racks less than 25 feet (300 inches) in height for inside warehouse installation.

1.2 Classification. The storage racks shall be one type, and of the following styles and sizes as specified (see 6.2):

Style A - 54-inch wide opening for one 40- by 48-inch pallet. Rated shelf capacity of 3,500 pounds.

Style B - 108-inch wide opening for two 40- by 48-inch pallets, positioned side by side. Rated shelf capacity of 6,000 pounds.

Size 1 upright frame - 120 inches high
Size 2 upright frame - 138 inches high
Size 3 upright frame - 144 inches high
Size 4 upright frame - 168 inches high
Size 5 upright frame - 198 inches high
Size 6 upright frame - 246 inches high
Size 7 upright frame - 294 inches high

1.3 Part or Identifying Number (PIN). A specification-based part number for items classified in this section shall be constructed as shown in 6.5.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, AMC Packaging, Storage, and Containerization Center, ATTN: SDSTO-TE-S, Tobyhanna, PA 18466-5097, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A AREA PACK

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto cited in the solicitation.

SPECIFICATIONS

FEDERAL

NN-P-71 - Pallet, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial).

MILITARY

MIL-P-15011 - Pallets, Material Handling, Wood Post Construction, 4-Way Entry.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1190- Minimum Guidelines for Level C Preservation, Packing, and Marking.

(Unless otherwise indicated, copies of Federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

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AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

Manual of Steel Construction

Specification for the Design, Fabrication and Erection of Structural Steel for Buildings.

(Application for copies should be addressed to the American Institute of Steel Construction, 101 Park Avenue, New York, NY 10017.)

AMERICAN IRON AND STEEL INSTITUTE (AISI)

Light Gage Cold-Formed Steel Design Manual.

Specification for the Design of Cold-Formed Steel Structural Members

(Application for copies should be addressed to the American Iron and Steel Institute, 1000 16th NW, Washington, DC 20036-5705.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM).

ASTM A123	- Zinc (Hot Dip Galvanized) Coatings on Iron & Steel Products, Standard Specification For
ASTM D3951	- Standard Practice for Commercial Packaging

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

A3.0-85	- Welding Terms and Definitions, Standard

(Application for copies should be addressed to the American Welding Society, 33 West 39th Street, New York, NY 10018.)

ENVIRONMENTAL SCIENCES SERVICES ADMINISTRATION (ESSA)

Coast and Geodetic Survey Seismic Map

(Application for copies should be addressed to U.S. Geological Survey, Branch of Seismicity and Risk Analysis (d1), Building 25, Denver Federal Center, Denver, CO 80225.)

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RACK MANUFACTURERS INSTITUTE (RMI)

Commentary and Supplementary Information - Specification for the Design, Testing and Utilization of Industrial Steel Storage Racks, 1985 Edition.

Specification for Design, Testing and Utilization of Industrial Steel Storage Racks - 1985 Edition.

(Application for copies may be addressed to Rack Manufacturers Institute, 8720 Red Oak Boulevard, Suite 201, Charlotte, NC 28217.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other information services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this specification shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specified exemption has been obtained.

3. REQUIREMENTS

3.1 Standard product. The standard product shall include all necessary rack components sized and finished as specified herein to permit normal warehouse assembly.

3.1.1 First article. Unless otherwise specified prior to commencement of production (see 6.2), a preproduction sample of the finished commodity or one taken from the contractor's current inventory, either of which shall include all component parts, shall be made available for first article inspection by the contracting officer or his authorized representative for compliance to this specification (see 4.3). Approval of the first sample authorizes the commencement of production or shipment of standard commercial product from inventory, but does not relieve the contractor of the responsibility of complying with all other provisions of this specification. Production units shall not vary from the approved preproduction sample without written approval of the contracting officer.

3.2 Materials. All materials shall be as hereinafter specified. All materials shall be new, unused, and free of all imperfections that affect strength, appearance, or serviceability of the finished commodity. Material not specified herein for any component(s) may be a manufacturer's commercial material which allows the components to meet the performance requirements of this specification. All materials shall meet the applicable quality assurance provisions and test requirements of section 4.

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3.2.1 Steel. The steel used for the fabrication of the storage racks shall be in accordance with one of the American Society for Testing Materials designations for structural steel referenced in the Specification for the Design, Fabrication, and Erection of Structural Steel for Buildings, or one of the suitable ASTM designations for structural quality sheet or strip steel referenced in the Specification for the Design of Cold-Formed Steel Structural Members. The steel used shall be of an ASTM designation grade (when applicable) and of sufficient strength and mass to assure compliance with all test requirements of this specification. The steel used for upright frames and load beams shall have a minimum yield strength of 50,000 pounds per square inch. The steel used for corner protectors shall be angle steel.

3.2.2 Wire mesh. Mesh shall be minimum 2-gauge steel wire and minimum 14-gauge steel channel angle support with a rated shelf capacity of 3,500 pounds.

3.3 Design. The design of the storage racks shall provide for a standard upright frame depth of 40 inches to achieve compatibility with the standard military pallet size of 40 inches (length) by 48 inches (width). The storage rack system (see 3.8) shall include 40-inch depth upright frames, style A (54 inches wide) or style B (108 inches wide) load beams, to accommodate one or two pallets, respectively, on each shelf opening. The design of all steel load-bearing components, hot formed to shape, shall be in accordance with the Specification for the Design, Fabrication, and Erection of Structural Steel for Buildings, as published in the Steel Construction Manual of the American Institute of Steel Construction (AISC), except as specified herein. The design of all steel load-bearing components, cold formed to shape, shall be in accordance with the Specification for the Design of Light Gage Cold-Formed Steel Structural Members, as published in the Light Gage Cold-Formed Steel Design Manual of the AISI, except as specified herein. The design and fabrication of each load-bearing component shall assure that the component is capable of continuously sustaining the design load, as specified (see 1.2). Also, a safety factor of not less than 1.65 for the shelf unit (load beams and front-to-rear support) and 1.92 for the upright frame are required. Typical storage rack design and construction are shown in figure 1.

3.4 Construction. The construction of the storage racks shall be as specified in the subsequent paragraphs. The upright frames, load beams, front-to-rear supports, and back-to-back rigid row connectors shall be constructed to produce a rigid, stable storage rack system that will withstand all tests specified in 4.6. The rack components shall be capable of assembly in the designated storage activity from components supplied by the contractor. The components shall be capable of assembly, side-by-side, in a continuous single row, and be connected back-to-back in a double row.

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3.5 Welding. All welding shall be accomplished by the suitable welding process referenced in American Welding Society A3.0-85. All welds shall be free of blowholes, slag, flash, and spatter. All welded joints shall withstand the industry tests and the tests and inspections specified in section 4.

3.6 Safety. The requirements for elements of safety for items specified shall implement and conform to the safety standards issued by the Occupational Safety and Health Administration under the Secretary of Labor.

3.7 Earthquakes. Unless otherwise specified (see 6.2), the design and construction of the rack components shall meet the requirements of the Uniform Building Code to resist lateral seismic forces (earthquakes). The contracting officer will provide the seismic zone based on the Coast and Geodetic Survey Seismic Map as obtained from Environmental Sciences Services Administration (ESSA).

3.8 Details of component parts.

3.8.1 Upright frame. The upright frame shall be of the material specified in 3.2.1 and shall be of a 40-inch depth to be compatible with the 40- by 48-inch NN-P-71 pallet. The frame shall be of all welded construction and consist of two upright posts, diagonal and horizontal braces, and floor-bearing plates. Each upright frame shall have a minimum safe load capacity consistent with the size of the upright frame (see 1.2 and 4.6.1.1). The posts shall have a channel, box channel, rectangular, or square formation. The width of the post, facing the cross or storage aisle, shall not exceed 3 inches. The bottom of each post shall have a floor-bearing plate of not less than a 14-square inch area and employ not less than 11 gauge (0.1196-inch) steel that is securely arc-welded to the bottom of the post. The plate shall have an accessible 0.4375-inch diameter hole for anchoring to the floor. Each upright post shall have openings to readily accommodate two shelf beam connectors, end-to-end, at corresponding heights throughout the entire height range of the post openings. A single or double row of openings shall commence 4 inches above the bottom of the upright (adjacent to the floor-bearing plate) and continue upward to the top of the post using 2- or 3-inch increments. All openings of each post shall be identical to each other and identical to the openings of all other posts in the contract or order. The positioning of all corresponding openings of the posts shall not vary more than 0.09375 inch as measured from the bottom of the floor-bearing plates. The arrangement of the openings and the fastening devices of the shelf unit shall be such that the top shelf unit is flush, ±0.375 inch with the top of the post. The upright posts shall be braced by horizontal and diagonal braces as required or depicted in the Rack Manufacturers Institute's (RMI) Specification for Design, Testing and Utilization of Industrial Steel Storage Racks, 1985 Edition.

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The braces shall have angle, channel, box channel, rectangular, square, "X" or "Z" formation. The braces shall be designed to minimize flexural, torsional, or torsional-flexural buckling as discussed under the Effective Lengths section of the RMI Storage Racks Commentary. Bottom and top horizontal braces shall be located not more than 10 inches from the bottom or the top of the upright frame. Under any one contract, the beam-to-upright frames of styles A and B storage racks shall be interchangeable. The frames shall be of the highest capacity and shall conform to the tests specified in 4.6.

3.8.2 Shelf unit. The shelf unit for style A racks shall have a minimum safe load capacity of 3,500 pounds and shall consist of two 54-inch length load beams and two drop- or slip-in, front-to-rear supports. The shelf unit for style B racks shall have a minimum safe load capacity of 6,000 pounds, and shall consist of two 108-inch length load beams and four drop- or slip-in, front-to-rear supports. The height of the assembled shelf unit shall not exceed 5 inches and the depth (front-to-rear) shall be compatible with the depth of the upright frame. All shelf units shall pass the tests and requirements of RMI storage rack specification, Pallet Beam in Upright Frames Assembly Test.

3.8.2.1 Shelf beam. The width across the bearing surface of the shelf beam shall be not less than 1.625 inches. The step-type shelf beam shall be of one- or two-piece construction of the materiel specified in 3.2.1. The one-piece beam formation shall not have less than two 90° bends, and the two-piece beam formation shall not have less than four 90° bends. The beam formations shall be formed accurately and uniformly, and the pieces of the two-piece beam shall be joined neatly and firmly on top and bottom, by chain or staggered intermittent welds, spaced on centers not further than 12 inches, as measured on the line of the welds. Each end of the beams shall have a multiple prong slip- or joint-fit, beam-to-post connector in accordance with the RMI storage rack specification listed in 2.2. The connector shall be capable of transferring all loads and external forces from the shelf beam to the post without failure of the connection or damage to the post or connector. The connections shall be of sufficient number, size, and strength to withstand repeated loading and unloading of the rack, and to continually sustain the rated shelf load. The prongs shall engage with the openings of the post in such a manner as to prevent lateral rotation of the shelf beam, with each prong contributing to the support of the shelf unit. The beam shall conform to the tests as specified in 4.6.

3.8.2.2 Beam-to-post safety lock. Each beam-to-post connection shall have a safety lock which will lock the beam to the post. The design and manner of construction of the device shall prevent accidental disengagement of the beam-to-post connection during movement or shifting of the assembled rack.

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The safety lock may be an integral part of the shelf beam or may consist of additional components. The safety lock shall be engaged without the use of tools, and no part thereof shall obstruct or restrict the loading or unloading of the shelf units.

3.8.2.3 Front-to-rear supports. The slip- or drop-in, front-to-rear supports shall be of the material specified in 3.2.1. The supports shall have a ~~channel, box channel, hat section,~~ rectangular, or square formation with a minimum width of 2 inches. Each front-to-rear support shall have a minimum safe load capacity of 1,750 pounds. The supports shall positively locate on the shelf beams and the top of the supports shall be flush, ±0.0625 inch, with the top of the shelf beams. The supports shall attach and detach with the use of a soft hammer to provide a tight fit between the load beams and the front-to-rear supports. If supports with hangers or flange-type attachments are used, the hangers or flanges may extend beyond the sides of the shelf beams but not more than 0.1875 inch. The front-to-rear supports shall conform to the tests specified in 4.6.1.5. The 54-inch load beam shelf requires two front-to-rear supports. The 108-inch load beam shelf requires four front-to-rear supports.

3.8.3 Back-to-back rigid row ties. Unless otherwise specified (see 6.2), back-to-back tie(s) shall be furnished with the rack unit. The back-to-back ties shall be capable of joining the backs of the same style units together to form a double row arrangement. The ties may be of a fixed or adjustable type, but shall provide a 4- to 6-inch spacing between the connected posts. Each end of the tie shall firmly and securely attach to any post of the upright frames, near the lower and upper limits of the upright frame, by bolts, washers, and nuts. Each end shall be attached to the post at not less than two points. Alternatively, the attachment may be boltless, with not less than two points of attachment at each post, at not less than two locations near the lower and upper limits of the upright frame. The boltless attachment shall have a safety locking device to prevent accidental disengagement. The back-to-back tie shall be of a size and strength to withstand all forces applied to the connected rack units during normal loading and unloading operations, without deformation or failure.

3.8.4 Leveling shim. Unless otherwise specified (see 6.2), leveling shims shall be provided with the rack units. The shims shall be hot or cold rolled steel and of the same size as the floor-bearing plate. The shims shall have a hole corresponding to the hole in the floor-bearing plate, and shall anchor or lock with the plate when the unit is erected. Two leveling shims shall be provided with each upright frame ordered.

3.8.5 Wire mesh decking. When specified, wire mesh decking shall be provided (see 6.2). It shall be welded or bolted to the steel channel angle supports not permanently attached to the

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front-to-rear supports or load shelf beams. The decking shall be removable from the rack shelf. It shall have a maximum 2- by 2-inch diamond- or square-shaped meshwork. Entire rack shelf with decking shall be galvanized.

3.8.6 Corner protectors. When specified for Army use (see 6.2), each exposed corner of upright racks shall be provided with protective corner bumpers constructed of angle steel. The bumpers shall rise a minimum of 36 inches from the floor base of rack, and shall be a minimum of 3/16 inch thick by 3 inches by 3 inches. Corners shall be freestanding and not fastened or welded to the upright frame.

3.8.7 Dimensions. The dimensions of the racks and components shall be as specified herein. See 1.2 for height and width of racks.

3.8.7.1 Tolerances. Tolerances for any component dimension shall be ±0.03125 inch. Tolerances for overall rack depth, height, and width shall be ±0.09375 inch. These tolerances may be cumulative but not to the extent that critical, specified, or overall tolerances cannot be maintained or the components and subassemblies will not easily assemble.

3.9 Color and finish.

3.9.1 Paint finish. All surfaces suitable for painting shall be thoroughly cleaned to provide a clean, dry surface, free of all mill scale, oil, grease, dirt, rust, and other contaminants. If manufacturer's normal commercial practice is to use primer, then one coat of good quality base primer shall be applied, followed (when dry) with a finish coat of high-quality paint to assure complete coverage and durability of finish. Application of the primer shall be integrated with the cleaning process and shall be applied as soon as possible after cleaning. The primer shall be chemically compatible with the finish coat. The finish coat may be air dried or baked on and shall provide a smooth, even surface, free of runs, sags, or other defects.

3.9.2 Color of finish. Unless otherwise specified (see 6.2), the color of the rack components' finish shall be the manufacturer's commercial color for storage racks. When color is specified, upright frames and back-to-back rigid row ties shall be semigloss green, color number 24449 of FED-STD-595. Shelf beams and front-to-rear supports shall be semigloss orange, color number 22510 of FED-STD-595. Sample standard color chips may be obtained from the General Services Administration, Specification Section, 7th D Street, SW, Room 6662, Washington, DC 20407. Protective corner bumpers shall be painted safety yellow.

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3.9.3 Galvanized finish. Galvanized racks shall employ components that comply with ASTM A-123 for hot galvanized-dipped coating (see 6.2).

3.9.4 Other type of finish. Other components of the rack system, not normally painted by the manufacturer, shall be cadmium-, chromium-, or zinc-plated to prevent corrosion.

3.10 Assembly. The upright frames and the shelf units shall assemble and disassemble, throughout the height range of the post openings, easily and freely, without the use of bolts or tools other than a soft hammer. Adjacent shelf units shall also assemble and disassemble with the common upright frame, easily and freely, without the use of tools other than a soft hammer. An assembled rack of two upright frames and three shelf units shall be rigid, stable, and without deformation or skewing when each shelf unit is loaded with a test load or to its capacity. The rack shall assemble so that the upright frames are perpendicular to the shelf units. The shelf units shall be parallel to each other. The upright frames of style A and B racks shall assemble with either end of all shelf units of style A and B racks. The beam-to-post safety lock shall engage and disengage without the use of tools and shall be effective at any position within the range of the shelf unit attachment. The leveling shim shall anchor or lock with the floor-bearing plate in such a manner as not to separate during normal use of the rack.

3.11 Identification marking. The components of the rack shall be free of any manufacturer's identification markings.

3.12 Certifications. A contractor shall submit a statement of certification by a professional engineer registered in the state of component manufacture. The certification shall relate to the overall design, reflected by shop drawings and calculations as well as to the design, weight, gauge, dimensions, and other characteristics of each numbered component piece part. The certification shall state that such designs and numbered component piece parts will fulfill the Government-specified loading and other requirements. Additionally, the manufacturer shall certify that all delivered rack components are those for which the registered professional engineer has provided certification.

3.13 Workmanship. The workmanship exercised throughout the various operations of manufacturing shall produce a serviceable product acceptable under the requirements specified herein. There shall be no missing parts or components. All welds, edges, and corners shall be smooth and free of sharp projections and burrs. Formed parts shall be uniform, without cracks, fractures, and undesirable tears and holes. The openings and holes in the upright frames shall be accurate and uniform. All referenced

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dimensions shall be within specified tolerance. All moveable parts shall operate freely and function properly. Attention shall be given to pretreatment operations to assure that all surfaces are pretreated and the proper sequence of operation is maintained and completed. The degree of workmanship shall assure that components will assemble and disassemble as specified in 3.10 and the occurrence of defects will not exceed the acceptable quality level specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements. However, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 First article. First article sample is as specified in 3.1.1. It consists of two end frames and three shelf units and includes all parts thereof.

4.3.2 Visual inspection. Sample shall be inspected for conformance to all requirements of this specification (see 4.4.2).

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4.3.3 Tests. Each first article sample shall be tested as specified in 4.6.

4.4 Quality conformance inspection.

4.4.1 Sampling. For acceptance and inspection, sampling shall be performed in accordance with MIL-STD-105. The lot size shall be expressed in units of racks.

4.4.2 Visual inspection. Inspection shall be performed in accordance with the classification of defects specified in table I. Inspection levels and acceptable quality levels shall be expressed in defects per hundred units in accordance with MIL-STD-105 and as shown in table II.

4.4.3 Dimensional inspection. Inspection shall be made of racks for compliance with dimensions and tolerances as specified (see 3.8.7). Any dimension not within tolerance specified shall be classified as a defect. The inspections and acceptable quality levels shall be as shown in table II.

TABLE I. Classification of defects.

Examine	Defect	Classification	
		Major	Minor
Steel	Steel previously used or not new stock	X	
	Steel contains scale, rust, pits, fins, sharp edges, protrusions, burrs, or undesirable holes	X	
Welds	Welding method employed other than specified		X
	Welds burned through or cracked	X	
	Rough, sharp projections, flash, and spatter not removed	X	
Pretreatment	Pretreatment omitted, primer omitted when required, and area(s) of no film	X	
	Poor adhesion or easily marred	X	

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TABLE I. Classifications of defects - Continued.

Examine	Defect	Classification	
		Major	Minor
	Blistering or peeling, cracking and flaking	X	
	Finish contains runs, sags, area(s) of thin film, or separation of color		X
	Color of finish other than specified		X
	Thickness not as specified		X
Upright frames	Frame design not constructed as specified	X	
	Number of horizontal and diagonal braces less than minimum specified	X	
	Top and bottom horizontal braces not located as specified	X	
	Floor-bearing plate not attached or size is less than minimum specified	X	
	Hole missing, wrong size, or not accessible in floor-bearing plate		X
	Openings in post will not accommodate two shelf beam connectors end to end	X	
	Openings in posts missing or not spaced in increments specified	X	
	Leveling shim will not anchor or lock with floor-bearing plate		X
Shelf units	Not as specified in 3.8.2	X	

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TABLE I. Classifications of defects - Continued.

Examine	Defect	Classification	
		Major	Minor
	Assembled height exceeds 5 inches		X
	Shelf beam formation not uniform		X
	Safety lock or locks missing, ineffective, or inoperative	X	
	Front-to-rear supports will not securely attach to shelf beams	X	
Back-to-back rigid row ties	Incapable of providing 4-inch spacing between posts	X	
	Each end has less than 2 points of attachment to post		X
	Boltless attachment not equipped with safety locking device	X	
Assembly	Components will not assemble and disassemble without the use of tools other than a soft hammer		X
	Shelf units will not assemble with posts throughout the range of openings	X	
	Style A rack components (except shelf beams) will not assemble to components of style B racks	X	
	Shelf unit will not assemble with upright frames within top and bottom limits specified in 3.8.1		X

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TABLE I. Classification of defects - Continued.

Examine	Defect	Classification	
		Major	Minor
	Assembled rack is not rigid and stable when loaded with full rated capacity or test loads	X	
Workmanship	Missing part or component, except bolt, washer, or nut	X	
	Missing bolt, washer, or nut		X
	Malformed or damaged part	X	
	Burrs, sharp edges, or sharp projections on welds, corners, or edges	X	
	Undesirable holes or tears in parts		X
	Holes and openings not accurate or uniform		X

TABLE II. Inspection and acceptable quality levels.

Inspection	Inspection Level	Acceptable Quality Major	Level Total
For defects listed in table I	S-2	4.0	10.0
For defects listed in table I	S-2	One class	4.0

4.5 Inspection of packaging.

4.5.1 Sampling. Sampling for inspection shall be in accordance with MIL-STD-105.

4.5.2 Inspection. Inspection shall be visual in accordance with specifications given in 5.2 and 5.3.

4.6 Tests.

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4.6.1 Load test. The following load tests shall be performed using inspection level S-1 for the selection of the number of racks to be tested from any lot. The lot size shall be expressed in units of racks. Style A and B racks shall be subject to the load test. Each unit tested shall be a completely assembled rack of two uprights and shelf units located at 48-inch incremental heights. The shelf units may be attached to the upright frames at convenient heights, except the top shelf unit shall be located where it will be free of any load applied to the top of the upright frames. All loads referenced in the individual test shall have a tolerance of ±1/2 of 1 percent; however, the minimum allowable load(s) shall not be intentionally used for the tests. All tests shall be conducted in the presence of the contracting officer or his designated representative. The Government reserves the right to use additional equipment, information, or methods deemed necessary to determine compliance with the requirements of this specification. The sequence of the test may be as herein delineated or any sequence of test favorable to existing circumstances may be used. The contractor has the option of having all tests performed on the racks as initially set up for testing, or he may substitute new components for those that have passed their particular test, before performing the next test. In the event the substituted components fail while subsequent tests are being performed, the previously recorded satisfactory performance for those components shall be declared void. Test loads mechanically applied to the pallets may be utilized, provided this method does not preclude compliance with any other requirement in the tests specified herein.

4.6.1.1 Upright frame load test. A load consisting of the total upright load shown in table III shall be applied to the two posts of an upright frame. The load shall be applied axially and as evenly as possible to the posts. The load shall remain on the posts for a period of 1 hour. At the end of 1 hour, the upright frames shall be visually examined, with the load applied, for evidence of deformation, skewing, joint separation, metal rupture, or component failure. Upon evidence of any of the above-stated defects, the test shall be stopped and the rack unit shall be considered as failing to meet the requirements of this specification. In absence of any of the above defects, the load shall be increased to the upright load times 1.5 as shown in table III. This load shall be applied for 30 minutes and then the upright frames shall be re-examined with the load applied for evidence of the defects stated above. Evidence of any of the above stated defects shall be sufficient cause for rejection of the rack unit.

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TABLE III. Upright loads based on maximum shelf loads.

Upright size	Upright height (inches)	Maximum number of shelves (1)	Upright load (lbs)	Upright load x 1.5 (lbs)
Style A Rack				
1	120	2	7,000	10,500
2	138	2	7,000	10,500
3	144	3	10,500	15,750
4	168	3	10,500	15,750
5	198	4	14,000	21,000
6	246	5	17,500	26,250
7	294	6	21,000	31,500
Style B Rack				
1	120	2	12,000	18,000
2	138	2	12,000	18,000
3	144	3	18,000	27,000
4	168	3	18,000	27,000
5	198	4	24,000	36,000
6	246	5	30,000	45,000
7	294	6	36,000	54,000

Note. Maximum number of shelves is based on a vertical distance of 48 inches between shelf beams (top of higher beam to top of lower beam or floor).

4.6.1.2 Alternative load test for upright frames. A stub column load test as stated in the section titled, Stub Column Tests of RMI Specification for the Design, Testing, and Utilization of Industrial Steel Storage Racks - 1985 Edition, may be conducted on a metal post segment in lieu of the upright frame load test described in 4.6.1.1. The segment tested shall be cut from a production metal post and have identical cross section, material thickness, and all other physical characteristics to the posts being supplied in the upright frames. The test shall be conducted in a qualified materials laboratory and all details of

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the test equipment, test procedures used, load requirements, and test results shall be provided the contracting officer in an original, typewritten, signed report. The report shall be signed by a qualified professional engineer who attests to the validity of the test results as complying with the maximum derived post load from table III. (The maximum upright frame load from table III shall be divided by two to provide the maximum metal post load.)

4.6.1.3 Shelf unit - style A racks. A 3,500-pound load shall be distributed as uniformly as possible on a 40- by 48-inch wood pallet. One loaded pallet shall be placed on each shelf unit. The shelf units shall have the loads applied simultaneously for a period of 1 hour. The deflection of any shelf beam at midspan, under load, in excess of 0.300 inch shall be sufficient cause for rejection of the rack unit. The measurement shall be made with a measuring device or devices with a unit calibration which does not exceed 0.010 inch. The measurement may be accomplished by a dial indicator, with an acceptable calibration, and precision space blocks, working from precision ground reference surface(s). Upon evidence of shelf beam deflection in excess of 0.300 inch, joint separation, metal rupture, or component failure, the test shall be stopped and the rack unit shall be considered as failing to meet the requirements of this specification. In the absence of any defects, the load on each pallet shall be increased to 5,250 pounds. This load shall be applied for 30 minutes and then removed. The unloaded rack shall be visually and dimensionally examined for evidence of permanent deformation, skewing, joint separation, metal rupture, and component failure. Evidence of any defects shall be sufficient cause for rejection of the rack unit (see table I).

4.6.1.4 Shelf unit - style B racks. A 6,000-pound uniformly distributed load shall be equally divided and placed on two 40-by 48-inch wood pallets. Two such loaded pallets shall be placed on each shelf unit and applied simultaneously for a period of 1 hour. The deflection of any shelf beam, under load, in excess of 0.600 inch shall be sufficient cause for rejection of the rack unit. The measurement shall be as specified in 4.6.1.3. Upon evidence of shelf beam deflection in excess of 0.600 inch, joint separation, metal rupture, or component failure, the test shall be stopped and the rack unit shall be considered as having failed to meet the requirements of this specification. In the absence of any of the defects stated above, the load on each pallet shall be increased to 4,500 pounds. This load shall be applied for 30 minutes and then removed. The unloaded rack unit shall be visually and dimensionally examined for evidence of permanent deformation, joint separation, metal rupture, and component failure. Evidence of any of these defects shall be sufficient cause for rejection of the rack unit.

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4.6.1.5 Front-to-rear supports. A load of 3,500 pounds shall be applied to an assembled pair of front-to-rear supports on one shelf unit. The load shall be distributed as uniformly as possible over the entire length of the supports. Each pair of front-to-rear supports of style B racks may be tested separately or simultaneously. The load shall be applied for 1 hour, after which the load shall be removed and the supports visually examined for evidence of deformation, skewing, joint separation, metal rupture, or component failure. Upon evidence of any defects, the test shall be stopped and the rack unit shall be considered as having failed to meet the requirements of this specification. In the absence of any defects, the load shall be increased to 5,250 pounds, applied for 30 minutes, and then removed. The supports shall be re-examined for evidence of defects. Evidence of any shall be sufficient cause for rejection of the rack unit.

4.6.1.6 Pallets for test loads. The pallets used when performing the tests specified in 4.6.1.3 to 4.6.1.5 shall be of the size specified in 1.2 and shall be of any construction covered by NN-P-71 and MIL-P-15011.

4.7 Acceptance standard. If visual examination of the sample rack units reveals defects in excess of acceptable quality levels cited in table II, or the load tests show that workmanship or materials are defective in any respect or do not comply with specified requirements including the test criteria cited in 4.6.1.1 to 4.6.1.5, the contractor shall correct all defects to the satisfaction of the contracting officer or his designated representative. At the option of the contracting officer or his designated representative, the load tests may be repeated after corrections have been made.

4.8 Final acceptance. Final acceptance shall be at destination contingent upon successful compliance with all quality assurance requirements and tests.

5. PACKAGING

5.1 Preservation. Not applicable.

5.2 Packing. Unless otherwise specified (see 6.2), packing shall be in accordance with ASTM D 3951 or for Army-only level C packing in accordance with MIL-STD-1190, shall be used when specified. Entire stacks of like rack components shall be banded with the steel strapping commonly used for that purpose. For handling by forklift or sling, 3- by 4-inch wooden runners shall be provided. Staples shall be used to position the strapping to the runners.

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5.3 Marking. In addition to any special or other identification marking required by the contract (see 6.2), all marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The boltless racks covered by this specification are intended for inside storage of palletized loads, and should allow placement and removal of any individual pallet without disturbing any other pallet(s).

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Style and size of rack units required (see 1.2).

c. When first article sample is not required (see 3.1.1).

d. Activity or installation location for determination of seismic (earthquake) load if required (see 3.7).

e. When back-to-back rigid row ties are not required (see 3.8.3).

f. Whether a spacing of 4- or 6-inches between connected posts is required (see 3.8.3).

g. When leveling shims are not required (see 3.8.4).

h. When wire mesh decking is required (see 3.8.5).

i. When corner protectors are required specify number (see 3.8.6).

j. When rack component's color, other than specified, is required (see 3.9.2).

k. When galvanized rack components are required (see 3.9.3).

l. When other than commercial packaging or military level C packaging is required (see 5.2).

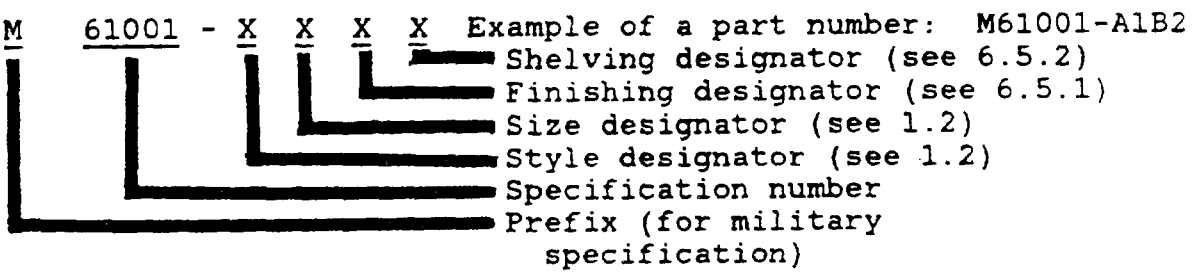
m. When special or other identification marking is required (see 5.3).

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6.3 First article. When first article inspection is required, the item should be a preproduction sample or if a standard production item, a sample from the contractor's current inventory may be made available (see 3.1 and 4.3.1). The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Supersession data. This specification supersedes and includes the requirements of interim Federal specification AA-R-0045 (GSA-FSS), dated 27 January 1969.

6.5 Part or identifying number. The part numbers to be used for the items covered in this specification are created as follows:



6.5.1 Finish designator. A one position field (single letter) shall be used to designate the desired finish. The regular finish (manufacturer's commercial color) shall be designated A, and the galvanized finish shall be designated B.

6.5.2 Shelving designator. A one position (single-digit number) shall be used to designate whether or not wire mesh shelving is wanted. The number 1 shall be used for a rack without shelving; the number 2 shall be used for a rack with shelving.

6.6 Subject term (key word) listing.

Storage rack, steel pallet
Storage racks, galvanized steel
Wire mesh shelves

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6.7 Changes from interim specification. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Miscellaneous. This specification does not include all types, styles, and sizes of rack units manufactured. The boltless rack units covered by this specification are of a style and size that meet the present requirements of the DOD.

Custodians:
Army - SM
Navy - SA
Air Force - 69

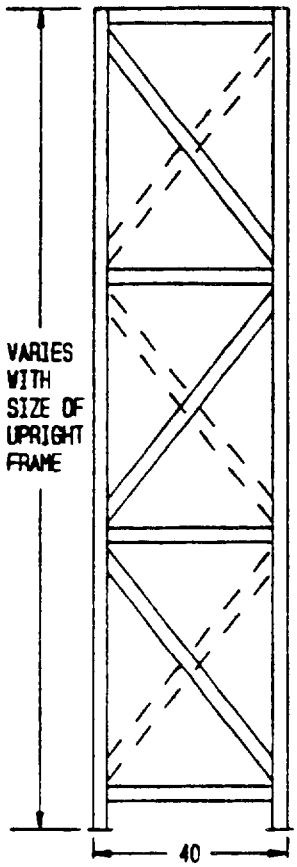
Review activities:
Army - AL, AR, TS,
Navy - OS
Air Force - 99
DLA - GS

Preparing activity:
Army - SM

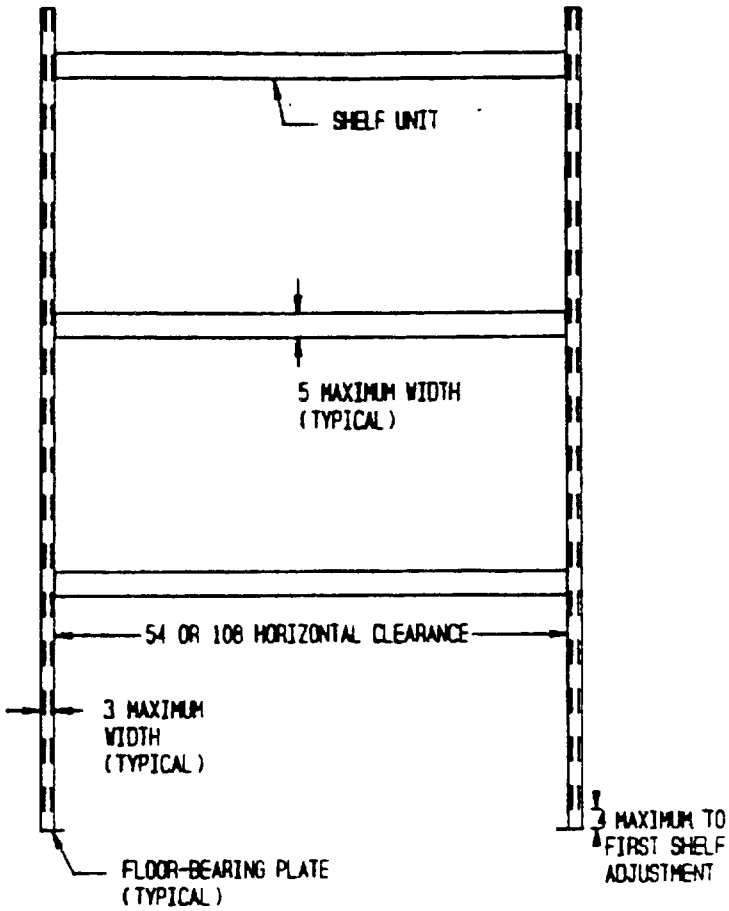
(Project PACK 0884)

Civil Agencies Coordinating Activities:
GSA-FSS

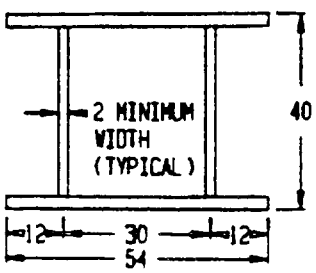
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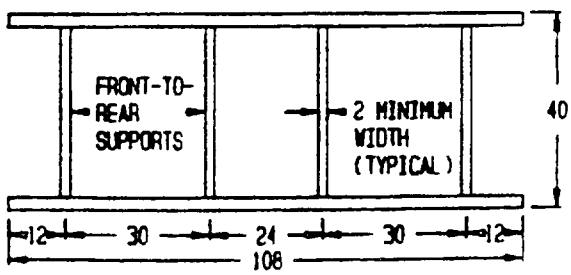
TYPICAL UPRIGHT FRAME CONSTRUCTION



ASSEMBLED STORAGE RACK



STYLE A SHELF UNIT



STYLE B SHELF UNIT

NOTE: ALL DIMENSIONS GIVEN ARE IN INCHES.

FIGURE 1. Racks, storage, loaded pallet (typical design and construction of).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-61001	Racks, Storage, Loaded Pallet

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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